

STAT 254 Problems Set

Set 3

1. An experiment consists of tossing a die and then flipping a coin once if the number on the die is even. If the number on the die is odd, the coin is flipped twice. Using the notation 4H, for example, to denote the outcome that the die comes up 4 and then the coin comes up heads, and 3HT to denote the outcome that the die comes up 3 followed by a head and then a tail on the coin, construct a tree diagram to show the 18 elements of the sample space S .

$$S = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}$$

2. If $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $A = \{0, 2, 4, 6, 8\}$, $B = \{1, 3, 5, 7, 9\}$, $C = \{2, 3, 4, 5\}$, and $D = \{1, 6, 7\}$, list the elements of the sets corresponding to the following events:

- (a) $A \cup C$;
- (b) $A \cap B$;
- (c) C' ;
- (d) $(C' \cap D) \cup B$;
- (e) $(S \cap C)'$;
- (f) $A \cap C \cap D'$.

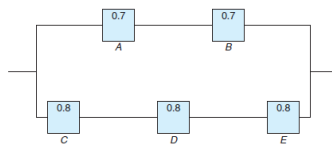
- (a) $\{0, 2, 3, 4, 5, 6, 8\}$
- (b) ϕ , the null set
- (c) $\{0, 1, 6, 7, 8, 9\}$
- (d) $\{1, 3, 5, 6, 7, 9\}$
- (e) $\{0, 1, 6, 7, 8, 9\}$
- (f) $\{2, 4\}$

Set 4

1. The probability that an American industry will locate in Shanghai, China, is 0.7, the probability that it will locate in Beijing, China, is 0.4, and the probability that it will locate in either Shanghai or Beijing or both is 0.8. What is the probability that the industry will locate
 - (a) in both cities? (0.3)
 - (b) in neither city? (0.2)
2. In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that
 - (a) the student took mathematics or history; (22/25)
 - (b) the student did not take either of these subjects; (3/25)
 - (c) the student took history but not mathematics. (17/50)

Set 5 - Set 7

1. The probability that a doctor correctly diagnoses a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient files a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues? 0.27
2. A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.
 - (a) What is the probability that neither is available when needed? 0.0016
 - (b) What is the probability that a fire engine is available when needed? 0.9984
3. A circuit system is given in Figure. Assume the components fail independently.



- (a) What is the probability that the entire system works? 0.75112
 - (b) Given that the system works, what is the probability that component A is not working? 0.2045
4. In a certain assembly plant, three machines, B1, B2, and B3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected.
 - (a) What is the probability that it is defective? 0.0245
 - (b) If a product was chosen randomly and found to be defective, what is the probability that it was made by machine B3? 0.204
 5. A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex? 0.857
 6. In a certain region of the country it is known from past experience that the probability of selecting an adult over 40 years of age with cancer is 0.05. If the probability of a doctor correctly diagnosing a person with cancer as having the disease is 0.78 and the probability of incorrectly diagnosing a person without cancer as having the disease is 0.06, what is the probability that an adult over 40 years of age is diagnosed as having cancer? 0.0960

Set 8 - Set 10

1. The probability distribution of X , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

- (a) Construct the cumulative distribution function of X .
 (b) $P(X \geq 2)$ 0.22
 (c) $P(X < 3)$ 0.94
 (d) $P(X > 1 | X < 3)$ 0.1702
 (e) $E(X)$ and $V(X)$ 0.88 and 0.8456
2. Determine the value c so that each of the following functions can serve as a probability distribution of the discrete random variable X : $f(x) = c(x^2 + 4)$, for $x = 0, 1, 2, 3$;
 (a) $C = 1/30$
 (b) $E(X)$ and $V(X)$ 2 and 1.13
 (c) $E(4X+3)$ 11
 (d) $V(-5X-3)$ 28.333
3. An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T , the number of years to maturity for a randomly selected bond, is

T	1	2	3	4
$F(t)$	0.25	0.5	0.8	1

- (a) Find $P(T=2)$. 0.25
 (b) $P(T \geq 3)$. 0.5
 (c) $P(T \geq 2 | T \leq 3)$. 0.6875
 (d) $E(X)$ and $V(X)$ 2.45 and 1.1475
 (e) $E(5-3X)$, $E(X^2)$ and $V(4+2X)$ -2.35, 7.15 and 4.59
4. Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period. 12.67

Set 11

1. It is conjectured that an impurity exists in 30% of all drinking wells in a certain rural community. In order to gain some insight into the true extent of the problem, it is determined that some testing is necessary. It is too expensive to test all of the wells in the area, so 10 are randomly selected for testing.
(a) Using the binomial distribution, what is the probability that exactly 3 wells have the impurity, assuming that the conjecture is correct? 0.2668
(b) What is the probability that more than 3 wells are impure? 0.3504
2. In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that
 - (a) from 3 to 6 have blowouts; 0.7073
 - (b) fewer than 4 have blowouts; 0.4613
 - (c) more than 5 have blowouts. 0.1484
3. Suppose that for a very large shipment of integrated-circuit chips, the probability of failure for any one chip is 0.10. Assuming that the assumptions underlying the binomial distributions are met, find the probability that at most 3 chips fail in a random sample of 20. (0.8670)

Set 12

1. An inventory study determines that, on average, demands for a particular item at a warehouse are made 5 times per day. What is the probability that on a given day this item is requested
 - (a) more than 5 times? 0.3840
 - (b) not at all? 0.0067
2. On average, a textbook author makes two word processing errors per page on the first draft of her textbook. What is the probability that on the next page she will make
 - (a) 4 or more errors? 0.1429
 - (b) no errors? 0.1353
3. The number of customers arriving per hour at a certain automobile service facility is assumed to follow a Poisson distribution with mean $\lambda = 7$.
 - (a) Compute the probability that more than 10 customers will arrive in a 2-hour period. 0.8243
 - (b) What is the mean number of arrivals during a 2-hour period? 14
4. Hospital administrators in large cities anguish about traffic in emergency rooms. At a particular hospital in a large city, the staff on hand cannot accommodate the patient traffic if there are more than 10 emergency cases in a given hour. It is assumed that

patient arrival follows a Poisson process, and historical data suggest that, on the average, 5 emergencies arrive per hour.

- (a) What is the probability that in a given hour the staff cannot accommodate the patient traffic? 0.0137
- (b) What is the probability that more than 20 emergencies arrive during a 3-hour shift? 0.0830

Set 13 and Set 14

1. The daily amount of coffee, in liters, dispensed by a machine located in an airport lobby is a random variable X having a continuous uniform distribution with $A = 7$ and $B = 10$. Find the probability that on a given day the amount of coffee dispensed by this machine will be

- (a) at most 8.8 liters; 0.6
- (b) more than 7.4 liters but less than 9.5 liters; 0.7
- (c) at least 8.5 liters. 0.5

2. Suppose that the error in the reaction temperature, in $^{\circ}\text{C}$, for a controlled laboratory experiment is

a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
- (b) Find $P(0 < X \leq 1)$. 1/9

$$F(x) = \begin{cases} 0, & x < -1, \\ \frac{x^3+1}{9}, & -1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

- (c) Find cdf $F(x)$ and use the cdf to Find $P(0 < X \leq 1)$.

4. The total number of hours, measured in units of 100 hours, that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that over a period of one year, a family runs their vacuum cleaner (a) less than 120 hours; 0.68

- (b) between 50 and 100 hours. 0.375

5. Consider the density function

$$f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Evaluate k. $3/2$

(b) Find $F(x)$ and use it to evaluate $P(0.3 < X < 0.6)$.

$$F(x) = \begin{cases} 0, & x < 0 \\ x^{3/2}, & 0 \leq x < 1; 0.3004 \\ 1, & x \geq 1 \end{cases}$$

6. Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20,000}{x^3}, & x > 100, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected life of this type of device. 200

7. The weekly demand for a drinking-water product, in thousands of liters, from a local chain of efficiency stores is a continuous random variable X having the probability density

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of X .

$5/3$ and $1/18$

8. The length of time, in minutes, for an airplane to obtain clearance for takeoff at a certain airport is a random variable $Y = 3X - 2$, where X has the density function

$$f(x) = \begin{cases} \frac{1}{4}e^{-x/4}, & x > 0 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of the random variable Y .

10 and 144

1. A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,
 - (a) what fraction of the cups will contain more than 224 milliliters?
 - (b) what is the probability that a cup contains between 191 and 209 milliliters?
 - (c) how many cups will probably overflow if 230- milliliter cups are used for the next 1000 drinks?
 - (d) below what value do we get the smallest 25% of the drinks?

(a) 0.0548; (b) 0.4514; (c) 23 cups; (d) 189.95 milliliters

2. A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live
 - (a) more than 32 months;
 - (b) less than 28 months;
 - (c) between 37 and 49 months.

(a) 0.8980; (b) 0.0287; (c) 0.6080

3. The probability that a patient recovers from a delicate heart operation is 0.9. Of the next 100 patients having this operation, what is the probability that
 - (a) between 84 and 95 inclusive survive?
 - (b) fewer than 86 survive?

(a) 0.9514; (b) 0.0668 with correction factor.

4. The average life of a certain type of small motor is 10 years with a standard deviation of 2 years. The manufacturer replaces free all motors that fail while under guarantee. If she is willing to replace only 3% of the motors that fail, how long a guarantee should be offered? Assume that the lifetime of a motor follows a normal distribution. 6.24

1. The length of time for one individual to be served at a cafeteria is a random variable having an exponential distribution with a mean of 4 minutes. What is the probability that a person is served in less than 3 minutes on at least 4 of the next 6 days?

$$\sum_{x=4}^6 \binom{6}{x} (1 - e^{-3/4})^x (e^{-3/4})^{6-x} = 0.3968$$

2. The length of time between breakdowns of an essential piece of equipment is important in the decision of the use of auxiliary equipment. An engineer thinks that the best model for time between breakdowns of a generator is the exponential distribution with a mean of 15 days.

(a) If the generator has just broken down, what is the probability that it will break down in the next 21 days? 0.7534

(b) What is the probability that the generator will operate for 30 days without a breakdown? 0.1353

- Let X be a random variable defined on an experiment, where X has probability distribution function:

X	1	3
$f(x)$	0.7	0.3

Suppose this experiment is conducted **twice**, and the average of the two experiments, $\bar{X}$, is calculated. Find the probability distribution table of $\bar{X}$.

- Suppose that 500 apples are ordered and you know from previous orders that the mean of an apple is 0.2 kilograms with standard deviation 0.1 kilograms. What is the probability that the total weight of the shipment is less than 98 kilograms?
- The average life of a bread-making machines is 7 years, with a standard deviation of 1 year. Assuming that the lives of these machines follow approximately a normal distribution, find
 - The probability that the mean life of a random sample of 9 such machines falls between 6.4 and 7.2 years;
 - The value of K to the right of which 15% of the means computed from random samples of sizes 9 would fall. Find K such that $P(\bar{X} \geq K) = 0.15$.
- The amount of time that a drive-through bank teller spends on a customer is a random variable with a mean 3.2 minutes and a standard deviation 1.6 minutes. If a random sample of 64 customers is observed, find the probability that their mean time at the teller's window is
 - At most 2.7 minutes;
 - at least 2.7 minutes but less than 3.4 minutes
- A farmer picks pumpkin from a large field. Suppose that 15% of all pumpkins is not fit to market in this field. The farmer makes samples of 260 pumpkins and inspects them. What is the probability that no more than 25% of this sample will be not fit to market?
- The mean of the sampling distribution of sample proportions has a value of p of 0.3, and sample size of 40. What is the probability that sample proportion is greater than 0.4

Answers:

- Twice: X_1 and X_2 are from the distribution of X (the table above)

$\bar{X}=(X_1+X_2)/2$ the distribution table is

$\bar{X}$	1	2	3
$f(\bar{x})$	0.49	0.42	0.09

2. 0.1867

Hint: $P(T < 98) = P(T/n < 98/500) = P(\bar{X} < 0.196)$

3. a) 0.6898 b) 7.35

4. a) 0.0062 b) 0.8351

5. 1

6. 0.0838

Set 19 – Set 21:

State the hypotheses:

- 1) The examinations in a large freshman chemistry class are scaled after grading so that the mean score is 75. The professor thinks that students who attend morning recitation sections will have a higher mean score than the class as a whole. Her students this semester can be considered a sample from the population of all students she might teach, so she compares their mean score with 75.
- 2) Last year, your company's service technicians took an average of 2.6 hours to respond to trouble calls from business customers who had purchased service contracts. Do this year's data show a different average response time?
- 3) In your weekly exercise routine, you usually run an average of 5.4 km per day. By the end of most weeks, you are usually quite tired and you suspect that the mean distance run on Fridays is lower than other days.

Complete the Hypothesis Test:

- 1) The level of calcium in the blood in healthy young adults varies with mean 9.5 mg/dL and standard deviation 0.4 mg/dL. A clinic in rural Guatemala measures the blood calcium level of 160 pregnant women at their first prenatal visit. Their mean is 9.57 mg/dL. Is this an indication that the mean calcium level differs from 9.5 in these women? Use $\alpha=5\%$.
Find a 95% confidence interval for the mean calcium level of pregnant women at their first prenatal visit.

P-value=0.0272, reject H_0

(9.508, 9.632)

- 2) A shop sells coffee where the number of pounds of coffee sold in a week is normal with mean, 320 and standard deviation, 40. After some advertising, the average 350 pounds of coffee is sold in the following 4 weeks. Has advertising improved business? Use $\alpha=5\%$.

Find a 95% confidence interval for the mean of coffee sold in a week.

P-value=0.0668, do not reject H_0

(280.8, 359.2)

- 3) A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

P-value= 0.0217, reject H_0

- 4) A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that $\mu = 8$ kilograms against the alternative that $\mu \neq 8$ kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.

P-value=0.0046

Set 22

- 1) Suppose that an engineer is interested in testing the bias in a pH meter. Data are collected on a neutral substance (pH = 7.0). A sample of the measurements is taken, with the data as follows:
- a. 7.00 7.10 6.97 7.00 7.03 7.01 7.01 6.98 7.08

It is, then, of interest to test

$$H_0: \mu = 7.0,$$

$$H_1: \mu \neq 7.0.$$

p-value>0.2

- 2) Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the contents of a random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal.

$t_{\text{obs}}=0.77$ p-value >0.2

- 3) Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a mean of 35 minutes. If a random sample of 20 high school seniors took an average of 33.1 minutes to complete this test with a standard deviation of 4.3 minutes, test the hypothesis, at the 0.05 level of significance, that $\mu = 35$ minutes against the alternative that $\mu < 35$ minutes.

$t_{\text{obs}}=-1.98$ $0.025 < \text{p-value} < 0.05$

Set 23

- 1) A builder claims that heat pumps are installed in 70% of all homes being constructed today in the city of Richmond, Virginia. Would you agree with this claim if a random survey of new homes in this city showed that 8 out of 15 had heat pumps installed? Use a 0.10 level of significance. P-value=0.1586
- 2) A marketing expert for a pasta-making company believes that 40% of pasta lovers prefer lasagna. If 9 out of 20 pasta lovers choose lasagna over other pastas, what can be concluded about the expert's claim? Use a 0.05 level of significance. P-value=0.6456

Set 24 and Set 25

- 1) A random sample of size $n_1 = 25$, taken from a normal population with a standard deviation 5.2, has a mean 81. A second random sample of size $n_2 = 36$, taken from a different normal population with a standard deviation 3.4, has a mean 76. Test the hypothesis that $\mu_1 = \mu_2$ against the alternative, $\mu_1 \neq \mu_2$. Z-test
- 2) An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. Can we conclude at the 0.05 level of significance that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units?
Pooled t $t_{\text{obs}}=1.04$ p-value>0.1
- 3) A manufacturer claims that the average tensile strength of thread A exceeds the average tensile strength of thread B by at least 12 kilograms. To test this claim, 50 pieces of each type of thread were tested under similar conditions. Type A thread had an average tensile strength of 86.7 kilograms with a standard deviation of 6.28 kilograms, while type B thread had an average tensile strength of 77.8 kilograms with a standard deviation of 5.61 kilograms. Test the manufacturer's claim using a 0.05 level of significance.
Z test $z_{\text{obs}}=-2.60$ $H_1: \mu_1 - \mu_2 < 12$ p-value=1-0.9953=0.0047

- 4) Engineers at a large automobile manufacturing company are trying to decide whether to purchase brand A or brand B tires for the company's new models. To help them arrive at a decision, an experiment is conducted using 12 of each brand. The tires are run until they wear out. The results are as follows:

Brand A : $\bar{x}_1 = 37,900$ kilometers, $s_1 = 4500$ kilometers.

Brand B : $\bar{x}_2 = 39,800$ kilometers, $s_2 = 6900$ kilometers.

Is there reason to believe that there is no difference in the average wear of the two brands of tires. Assume the populations to be approximately normally distributed with equal variances.

$$p\text{-value} > 0.2$$

Set 26

- 1) A taxi company manager is trying to decide whether the use of radial tires instead of regular belted tires improves fuel economy. Twelve cars were equipped with radial tires and driven over a prescribed test course. Without changing drivers, the same cars were then equipped with regular belted tires and driven once again over the test course. The gasoline consumption, in kilometers per liter, was recorded as follows:

Car	Kilometers per Liter	
	Radial Tires	Belted Tires
1	4.2	4.1
2	4.7	4.9
3	6.6	6.2
4	7.0	6.9
5	6.7	6.8
6	4.5	4.4
7	5.7	5.7
8	6.0	5.8
9	7.4	6.9
10	4.9	4.7
11	6.1	6.0
12	5.2	4.9

Can we conclude that cars equipped with radial tires give better fuel economy than those equipped with belted tires? Assume the populations to be normally distributed.

$$0.01 < p\text{-value} < 0.025$$

- 2) Five samples of a ferrous-type substance were used to determine if there is a difference between a laboratory chemical analysis and an X-ray fluorescence analysis of the iron content. Each sample was split into two subsamples and the two types of analysis were applied. Following are the coded data showing the iron content analysis:

Analysis	Sample				
	1	2	3	4	5
X-ray	2.0	2.0	2.3	2.1	2.4
Chemical	2.2	1.9	2.5	2.3	2.4

Assuming that the populations are normal, test at the 0.05 level of significance whether the two methods of analysis give, on the average, the same result.

$$0.2 > p\text{-value} > 0.1$$

Set 27

- 1) In a study to estimate the proportion of residents in a certain city and its suburbs who favor the construction of a nuclear power plant, it is found that 63 of 100 urban residents favor the construction while only 59 of 125 suburban residents are in favor. Is there a significant difference between the proportions of urban and suburban residents who favor construction of the nuclear plant? $P\text{-value}=0.0164$
- 2) A vote is to be taken among the residents of a town and the surrounding county to determine whether a proposed chemical plant should be constructed. The construction site is within the town limits, and for this reason many voters in the county believe that the proposal will pass because of the large proportion of town voters who favor the construction. To determine if there is a significant difference in the proportions of town voters and county voters favoring the proposal, a poll is taken. If 120 of 200 town voters favor the proposal and 240 of 500 county residents favor it, would you agree that the proportion of town voters favoring the proposal is higher than the proportion of county voters? $P\text{-value}=0.0018$

Set 28- Set 30

- 1) Physical properties of six flame-retardant fabric samples were investigated in a report. Use the accompanying data

Stiffness (x)mg-cm	7.98	24.52	12.47	6.92	24.11	35.71
Thickness (y) mm	0.28	0.65	0.32	0.27	0.81	0.57

- a) Compute the covariance of X and Y, the correlation coefficient and coefficient of determination.
- b) Estimate the slope and intercept through the data using least squares regression.
- c) What is the residual at stiffness (x) 6.92?
- d) What is the predicted value (fitted value) of Thickness (y), using the simple linear regression model, when the stiffness (x) is 20.
- e) Use a 0.05 significance level to determine whether a linear relationship exists between x and y.
 - 1) Test the parameter of slope, β_1 .
 - 2) Test the coefficient of correlation, ρ .

2). Plasma etching is essential to the fine-line pattern transfer in semiconductor process. The following table shows that data chlorine flow (x, in SCCM) through a nozzle used in the etching mechanism and etch rate (y, in 100 Å/min).

X	1.5	1.5	2.0	2.5	2.5	3.0	3.5	3.5	4.0
y	23.0	24.5	25.0	30.0	33.5	40.0	40.5	47.0	49.0

- a). Compute the covariance of X and Y, the correlation coefficient and coefficient of determination.
- b.) Estimate the slope and intercept through the data using least squares regression.
- c.) What is the residual at x=4.0?
- d.) What is the predicted value (fitted value) of etch rate (y), using the simple linear regression model, when the chlorine flow (x) is 2.7.
- e.) Compute the SSE.
- f.) Use a 0.05 significance level to determine whether a linear relationship exists between x and y.
 - 1.) Test the parameter of slope, β_1 .
 - 2.) Test the coefficient of correlation, ρ .

3). Medical researchers have noted that adolescent females are much more likely to deliver low-birth-weight babies than are adult females. Because such babies have higher mortality rates, numerous investigations have focused on the relationship between mother's age and birth weight. One report has the following data on x=maternal age (year) and y=baby's birth weight (g):

X	15	17	18	15	16	19	17	16	18	19
y	2289	3393	3271	2648	2897	3327	2970	2535	3138	3573

- a). Compute the covariance of X and Y, the correlation coefficient and coefficient of determination.
- b). Estimate the slope and intercept through the data using least squares regression.
- c). What is the residual at the data (15, 2289)?
- d). What is the predicted value (fitted value) of baby's birth weight (y), using the simple linear regression model, when the maternal age (x) is 18.5.
- e). Compute the SSE.
- f). Use a 0.05 significance level to determine whether a linear relationship exists between x and y.

- 1.) Test the parameter of slope, β_1 .
- 2.) Test the coefficient of correlation, ρ .

Set 28 - 30
1.

Conclusion:

(a)

$$\begin{aligned} S_{xy} &= 1.984 \\ r &= 0.773 \\ r^2 &= 0.597 \end{aligned} \quad (2)$$

$\alpha = 0.05$: Do not reject H_0 .

(b)

$$\begin{aligned} b_0 &= 0.1969 \\ b_1 &= 0.0154 \end{aligned}$$

$$\begin{aligned} H_0 : \rho &= 0 \\ H_1 : \rho &\neq 0 \end{aligned}$$

(c)

$$\begin{aligned} \hat{y} &= 0.1969 + 0.0154X \\ \hat{y} &= 0.1969 + 0.0154(6.92) = 0.30 \\ e &= y - \hat{y} \\ e &= 0.27 - 0.30 = -0.03 \end{aligned} \quad \begin{matrix} 2. \\ (a) \end{matrix}$$

$$\begin{aligned} t_{\text{obs}} &= r \sqrt{\frac{n-2}{1-r^2}} = 2.436 \\ &\sim t_4 \end{aligned}$$

(d)

$$\hat{y} = 0.1969 + 0.0154(20) = 0.50$$

$$\begin{aligned} S_{xy} &= 8.6146 \\ r &= 0.9703 \\ r^2 &= 0.9415 \end{aligned}$$

(e)

$$\begin{aligned} H_0 : \beta &= 0 \\ H_1 : \beta &\neq 0 \end{aligned} \quad (b)$$

$$t_{\text{obs}} = \frac{0.0154}{S_{b1}} = 2.436$$

$$\begin{aligned} b_0 &= 6.449 \\ b_1 &= 10.6026 \end{aligned}$$

Key calculations:

(c)

$$\begin{aligned} S_{b1} &= \frac{S_e}{\sqrt{(n-1)S_{xx}}} \\ S_e &= \sqrt{\frac{\text{SSE}}{n-2}} = 0.1604 \\ \text{SSE} &= (n-1)(S_y^2 - S_{xy}^2/S_x^2) = 5(0.0226) = 0.1039 \end{aligned}$$

$$\begin{aligned} \hat{y} &= 6.449 + 10.6026 \cdot x \\ x &= 4.0 \\ \hat{y} &= 48.8594 \end{aligned}$$

P-value:

$$\begin{aligned} P\text{-value} &= 2P(t_{4>2.436}) \\ 0.025 * 2 &< P\text{-value} < 0.05 * 2 \\ 0.05 &< P\text{-value} < 0.1 \end{aligned}$$

$$\begin{aligned} e &= y - \hat{y} \\ e &= 49 - 48.8594 = 0.1406 \end{aligned}$$

$$\begin{aligned} \hat{y} &= 6.449 + 10.6026 \cdot 2.7 \\ \hat{y} &= 35.076 \end{aligned}$$

(e)

$$\text{SSE} = 45.3622$$

(f) 1. Hypothesis test:

$$\begin{aligned} H_0 : \beta &= 0 \\ H_1 : \beta &\neq 0 \end{aligned} \tag{e}$$

$$SSE = 3372125$$

$$\begin{aligned} t_{\text{obs}} &= 10.619 \sim t_7 \\ \text{P-value} &< 0.005 \end{aligned}$$

(f) 1. Hypothesis test:

Conclusion:

$$\begin{aligned} H_0 : \beta &= 0 \\ H_1 : \beta &\neq 0 \end{aligned}$$

$$\text{Reject } H_0$$

2. Hypothesis test:

$$\begin{aligned} t_{\text{obs}} &= 5.34 \sim t_8 \\ \text{P-value} &< 0.005 \end{aligned}$$

$$\begin{aligned} H_0 : \rho &= 0 \\ H_1 : \rho &\neq 0 \end{aligned}$$

Conclusion:

$$\text{Reject } H_0$$

$$\begin{aligned} t_{\text{obs}} &= 10.619 \sim t_7 \\ \text{P-value} &< 0.005 \end{aligned}$$

2. Hypothesis test:

Conclusion:

$$\begin{aligned} H_0 : \rho &= 0 \\ H_1 : \rho &\neq 0 \end{aligned}$$

$$\text{Reject } H_0$$

3.

$$\begin{aligned} t_{\text{obs}} &= 5.34 \sim t_8 \\ \text{P-value} &< 0.005 \end{aligned}$$

(a)

$$\begin{aligned} S_{xy} &= 544.7788 \\ r &= 0.8837 \\ r^2 &= 0.7809 \end{aligned}$$

Conclusion:

$$\text{Reject } H_0$$

(b)

$$\begin{aligned} b_0 &= -1163.45 \\ b_1 &= 245.15 \end{aligned}$$

(c)

$$e = -248.8$$

(d)

$$\begin{aligned} \hat{y} &= -1163.45 + 245.15 \cdot 18.15 \\ \hat{y} &= 3371.825 \end{aligned}$$

Set 31 and Set 32

- 1) A die is tossed 180 times with the following results:

x	1	2	3	4	5	6
f	28	36	36	30	27	23

Is this a balanced die? $X_{\text{obs}}=4.47$

- 2) A random sample of 90 adults is classified according to gender and the number of hours of television watched during a week:

	Gender	
	Male	Female
Over 25 hours	15	29
Under 25 hours	27	19

Use a 0.01 level of significance and test the hypothesis that the time spent watching television is independent of whether the viewer is male or female.

$X_{\text{obs}}=5.47$